

❖ Expense.java

```
package my.project;

import jakarta.persistence.*;

@Entity

public class Expense {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    private String category;
    private Double amount;
    private String description;
    public Expense() {}
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getCategory() { return category; }
    public void setCategory(String category) { this.category = category; }
    public Double getAmount() { return amount; }
    public void setAmount(Double amount) { this.amount = amount; }
    public String getDescription() { return description; }
    public void setDescription(String description) {
        this.description = description;
    }
    public void setAmount(double amount) {
        this.amount = amount;
    }
}
```

❖ ExpenseController.java

```
package my.project;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;

import jakarta.servlet.http.HttpServletResponse;
import java.util.List;

import com.itextpdf.kernel.pdf.*;
import com.itextpdf.layout.Document;
import com.itextpdf.layout.element.Paragraph;

@Controller
public class ExpenseController {

    @Autowired
    private ExpenseRepository expenseRepository;

    // =====
    // HOME PAGE
    // =====
    @GetMapping("/")
    public String home(Model model) {

        List<Expense> expenses = expenseRepository.findAll();

        double totalExpense = expenses.stream()
            .mapToDouble(Expense::getAmount)
            .sum();

        double salary = 50000;
        double balance = salary - totalExpense;

        model.addAttribute("expenses", expenses);
        model.addAttribute("total", totalExpense);
        model.addAttribute("salary", salary);
    }
}
```

```

        model.addAttribute("balance", balance);

        return "index";
    }

    // =====
    // ADD
    // =====
    @PostMapping("/add")
    public String addExpense(@ModelAttribute Expense expense) {
        expenseRepository.save(expense);
        return "redirect:/";
    }

    // =====
    // DELETE
    // =====
    @GetMapping("/delete/{id}")
    public String deleteExpense(@PathVariable Long id) {
        expenseRepository.deleteById(id);
        return "redirect:/";
    }

    // =====
    // EDIT
    // =====
    @GetMapping("/edit/{id}")
    public String editExpense(@PathVariable Long id, Model model) {
        Expense expense = expenseRepository.findById(id).orElse(null);
        model.addAttribute("expense", expense);
        return "edit";
    }

    // =====
    // UPDATE
    // =====
    @PostMapping("/update")
    public String updateExpense(@ModelAttribute Expense expense) {
        expenseRepository.save(expense);
        return "redirect:/";
    }

```

```

    }

    // =====
    // REPORT PAGE
    // =====
    @GetMapping("/report")
    public String generateReport(Model model) {

        List<Expense> expenses = expenseRepository.findAll();

        double totalExpense = expenses.stream()
            .mapToDouble(Expense::getAmount)
            .sum();

        double salary = 50000;
        double balance = salary - totalExpense;

        model.addAttribute("expenses", expenses);
        model.addAttribute("total", totalExpense);
        model.addAttribute("salary", salary);
        model.addAttribute("balance", balance);
        model.addAttribute("count", expenses.size());

        return "report";
    }

    // =====
    // DOWNLOAD PDF
    // =====
    @GetMapping("/download")
    public void downloadReport(HttpServletResponse response) throws
Exception {

        response.setContentType("application/pdf");
        response.setHeader("Content-Disposition",
            "attachment; filename=TrackWise_Report.pdf");

        List<Expense> expenses = expenseRepository.findAll();

        double total = expenses.stream()

```

```

        .mapToDouble(Expense::getAmount)
        .sum();

double salary = 50000;
double balance = salary - total;
PdfWriter writer = new PdfWriter(response.getOutputStream());
PdfDocument pdf = new PdfDocument(writer);
Document document = new Document(pdf);
document.add(new Paragraph("TrackWise Monthly Report"));
document.add(new Paragraph(" "));
document.add(new Paragraph("Salary: ₹" + salary));
document.add(new Paragraph("Total Expense: ₹" + total));
document.add(new Paragraph("Remaining Balance: ₹" + balance));
document.add(new Paragraph(" "));
document.add(new Paragraph("Expense List:"));
document.add(new Paragraph("-----"));
for (Expense e : expenses) {
    document.add(new Paragraph(
        e.getDescription() + " - ₹" + e.getAmount()));
}
document.close();
}
}

```

❖ ExpenseRepository.java

```

package my.project;
import org.springframework.data.jpa.repository.JpaRepository;
public interface ExpenseRepository extends JpaRepository<Expense,
Long> {
}

```

❖ LoginController.java

```

package my.project;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
@Controller
public class LoginController {
    @GetMapping("/login")
    public String login() {

```

```

        return "login";
    }
}

```

❖ **SecurityConfig.java**

```

package my.project;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import
org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.web.SecurityFilterChain;
@Configuration
public class SecurityConfig {
    @Bean
    public SecurityFilterChain filterChain(HttpSecurity http) throws
Exception {

        http
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/login", "/css/**").permitAll()
                .anyRequest().authenticated()
            )
            .formLogin(form -> form
                .loginPage("/login")
                .defaultSuccessUrl("/", true)
                .permitAll()
            )
            .logout(logout -> logout
                .logoutSuccessUrl("/login?logout")
                .permitAll()
            );
        return http.build();
    }
}

```

❖ **TrackWiseApplication.java**

```

package my.project;
import org.springframework.boot.SpringApplication;

```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class TrackWiseApplication {
    public static void main(String[] args) {
        SpringApplication.run(TrackWiseApplication.class, args);
    }
}
```

❖ edit.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
    <title>Edit Expense</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #4facfe, #00f2fe);
            margin: 0;
            padding: 0;
        }

        h2 {
            text-align: center;
            color: white;
            padding: 20px;
        }

        .container {
            width: 40%;
            margin: auto;
            background: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0px 4px 15px rgba(0,0,0,0.2);
        }

        input {
            width: 100%;
            padding: 10px;
```

```
margin-top: 5px;
margin-bottom: 20px;
border-radius: 5px;
border: 1px solid #ccc;
}
```

```
button {
  width: 100%;
  padding: 10px;
  background-color: #4CAF50;
  color: white;
  border: none;
  border-radius: 5px;
  font-weight: bold;
  cursor: pointer;
}
```

```
button:hover {
  background-color: #45a049;
}
```

```
.back-btn {
  text-align: center;
  margin-top: 15px;
}
```

```
.back-btn a {
  text-decoration: none;
  color: #4facfe;
  font-weight: bold;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Edit Expense</h2>
```

```
<div class="container">
```



```

    <form th:action="@{/update}" th:object="${expense}"
method="post">

        <input type="hidden" th:field="*{id}" />
        <label>Description</label>
        <input type="text" th:field="*{description}" required>

        <label>Amount (₹)</label>
        <input type="number" step="0.01" th:field="*{amount}" required>

        <button type="submit">Update Expense</button>

    </form>

    <div class="back-btn">
        <a th:href="@{/}">← Back to Home</a>
    </div>

</div>

</body>
</html>

```

❖ index.html

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
    <title>TrackWise - Expense Tracker</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #4facfe, #00f2fe);
            margin: 0;
            padding: 0;
        }

        h1 {
            text-align: center;
            color: white;

```

```
padding: 20px;  
}
```

```
.container {  
  width: 80%;  
  margin: auto;  
  background: white;  
  padding: 25px;  
  border-radius: 10px;  
  box-shadow: 0px 4px 15px rgba(0,0,0,0.2);  
  margin-bottom: 40px;  
}
```

```
.summary {  
  background-color: #f9f9f9;  
  padding: 15px;  
  border-radius: 8px;  
}
```

```
input {  
  padding: 8px;  
  width: 250px;  
  border-radius: 5px;  
  border: 1px solid #ccc;  
}
```

```
button {  
  padding: 8px 15px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
.add-btn {  
  background-color: #4CAF50;  
  color: white;  
}
```

```
.report-btn {  
  background-color: #2196F3;
```

```
    color: white;  
}
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
}
```

```
th {  
    background-color: #2196F3;  
    color: white;  
    padding: 10px;  
}
```

```
td {  
    padding: 10px;  
    text-align: center;  
}
```

```
tr:nth-child(even) {  
    background-color: #f2f2f2;  
}
```

```
.edit-btn {  
    color: blue;  
    font-weight: bold;  
}
```

```
.delete-btn {  
    color: red;  
    font-weight: bold;  
}
```

```
.good {  
    color: green;  
    font-weight: bold;  
}
```

```
.bad {  
    color: red;
```

```

        font-weight: bold;
    }
</style>
</head>
<body>

<h1> 💰 TrackWise - Expense Tracker</h1>

<div class="container">

    <div class="summary">
        <h2>Budget Summary</h2>

        <p><b>Salary:</b> ₹<span th:text="\${salary}"></span></p>
        <p><b>Total Expense:</b> ₹<span th:text="\${total}"></span></p>
        <p><b>Remaining Balance:</b> ₹<span
th:text="\${balance}"></span></p>

        <div th:if="\${balance > 0}" class="good">
            🎉 Great! You saved ₹<span th:text="\${balance}"></span>
        </div>

        <div th:if="\${balance == 0}">
            ⚡ No savings this month.
        </div>

        <div th:if="\${balance < 0}" class="bad">
            ⚠ Expenses Exceeded Salary!
        </div>
    </div>

<hr>

<h2>Add Expense</h2>

<form th:action="@{/add}" method="post">
    <label>Description:</label><br>
    <input type="text" name="description" required><br><br>

```

```

        <label>Amount:</label><br>
        <input type="number" step="0.01" name="amount"
required><br><br>

        <button type="submit" class="add-btn">Add Expense</button>
    </form>

    <hr>

    <h2>All Expenses</h2>

    <table>
        <tr>
            <th>Description</th>
            <th>Amount (₹)</th>
            <th>Action</th>
        </tr>

        <tr th:each="expense : ${expenses}">
            <td th:text="${expense.description}"></td>
            <td th:text="${expense.amount}"></td>
            <td>
                <a th:href="@{/edit/{id}(id=${expense.id})}" class="edit-
btn">Edit</a> |
                <a th:href="@{/delete/{id}(id=${expense.id})}"
class="delete-btn"
onclick="return confirm('Delete this expense?');">
Delete
                </a>
            </td>
        </tr>
    </table>

    <br><br>

    <a th:href="@{/report}">
        <button class="report-btn"> 📄 Generate Final Report</button>
    </a>

</div>

```

```
</body>
```

```
</html>
```

❖ login.html

```
<!DOCTYPE html>
```

```
<html xmlns:th="http://www.thymeleaf.org">
```

```
<head>
```

```
<title>Login</title>
```

```
<style>
```

```
/* Full-page background */
```

```
body {
```

```
margin: 0;
```

```
height: 100vh;
```

```
display: flex;
```

```
justify-content: center; /* horizontal center */
```

```
align-items: center; /* vertical center */
```

```
font-family: Arial, sans-serif;
```

```
background: linear-gradient(135deg, #f6d365, #fda085);
```

```
}
```

```
/* Login form container */
```

```
.login-container {
```

```
background-color: rgba(255, 255, 255, 0.9);
```

```
padding: 40px;
```

```
border-radius: 15px;
```

```
box-shadow: 0 8px 20px rgba(0, 0, 0, 0.3);
```

```
width: 300px;
```

```
text-align: center;
```

```
}
```

```
h2 {
```

```
margin-bottom: 25px;
```

```
color: #333;
```

```
}
```

```
input[type="text"], input[type="password"] {
```

```
width: 90%;
```

```
padding: 10px;
```

```
margin: 10px 0;
```

```
border: 2px solid #fda085;
```

```
border-radius: 5px;
```

```
outline: none;
```

```
transition: 0.3s;
```

```

    }
    input[type="text"].focus, input[type="password"].focus {
        border-color: #f6d365;
    }
    button {
        background-color: #f6d365;
        border: none;
        padding: 12px 25px;
        border-radius: 5px;
        cursor: pointer;
        color: white;
        font-weight: bold;
        transition: 0.3s;
    }
    button:hover {
        background-color: #fda085;
    }
    .error {
        color: red;
        margin-bottom: 15px;
    }
</style>
</head>
<body>
    <div class="login-container">
        <h2>Login Page</h2>
        <form th:action="@{/login}" method="post">
            <input type="text" name="username"
placeholder="Username"/><br/>
            <input type="password" name="password"
placeholder="Password"/><br/>
            <div th:if="${param.error}" class="error">
                ✖ Wrong username or password!
            </div>
            <button type="submit">Login</button>
        </form>
    </div>
</body>
</html>

```

❖ report.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <title>Final Expense Report</title>

  <style>
    body {
      font-family: Arial;
      background: linear-gradient(to right, #4facfe, #00f2fe);
      padding: 20px;
    }

    .container {
      width: 60%;
      margin: auto;
      background: white;
      padding: 30px;
      border-radius: 10px;
      box-shadow: 0px 4px 15px rgba(0,0,0,0.2);
    }

    table {
      width: 100%;
      margin-top: 20px;
      border-collapse: collapse;
    }

    th {
      background: #2196F3;
      color: white;
      padding: 10px;
    }

    td {
      padding: 8px;
      border-bottom: 1px solid #ddd;
    }
  </style>
</head>
```



```
.good {  
  color: green;  
  font-weight: bold;  
}
```

```
.bad {  
  color: red;  
  font-weight: bold;  
}
```

```
button {  
  padding: 8px 15px;  
  border: none;  
  border-radius: 5px;  
  color: white;  
  cursor: pointer;  
}
```

```
.download-btn {  
  background-color: #f44336;  
}
```

```
.back-btn {  
  background-color: #4CAF50;  
}
```

```
</style>  
</head>  
<body>
```

```
<div class="container">
```

```
<h2>  TrackWise Monthly Report</h2>
```

```
<p><b>Salary:</b> ₹<span th:text="\${salary}"></span></p>  
<p><b>Total Expenses:</b> ₹<span th:text="\${total}"></span></p>  
<p><b>Remaining Balance:</b> ₹<span  
th:text="\${balance}"></span></p>  
<p><b>Total Transactions:</b> <span  
th:text="\${count}"></span></p>
```

```
<div th:if="{balance > 0}" class="good">
```

```
     Good Financial Health
```

```
</div>
```

```
<div th:if="{balance < 0}" class="bad">
```

```
     Expenses exceeded salary!
```

```
</div>
```

```
<table>
```

```
    <tr>
```

```
        <th>Description</th>
```

```
        <th>Amount (₹)</th>
```

```
    </tr>
```

```
    <tr th:each="expense : {expenses}">
```

```
        <td th:text="{expense.description}"></td>
```

```
        <td th:text="{expense.amount}"></td>
```

```
    </tr>
```

```
</table>
```

```
<br><br>
```

```
<a th:href="{/download}">
```

```
    <button class="download-btn">↓ Download PDF</button>
```

```
</a>
```

```
<a th:href="{/}">
```

```
    <button class="back-btn">← Back to Home</button>
```

```
</a>
```

```
</div>
```

```
</body>
```

```
</html>
```

❖ applications.properties

```
spring.datasource.url=jdbc:mysql://localhost:3306/trackwise
```

```
spring.datasource.username=root
```

```
spring.datasource.password=likhi9985
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.database-platform=org.hibernate.dialect.MySQL8Dialect
spring.thymeleaf.cache=false
server.port=9085
spring.security.user.name=admin
spring.security.user.password=admin123
```

❖ pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>my.project</groupId>
  <artifactId>trackwise</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <packaging>jar</packaging>
  <name>TrackWise</name>
  <description>Expense Tracker</description>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.2.5</version>
    <relativePath/>
  </parent>
  <properties>
    <java.version>17</java.version>
  </properties>
  <dependencies>
    <!-- Spring Boot Web -->
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <!-- Thymeleaf -->
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-thymeleaf</artifactId>
    </dependency>
  </dependencies>
</project>
```

```
</dependency>
<!-- Spring Data JPA -->
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
<!-- Spring Security (IMPORTANT FOR YOUR ERROR) -->
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-security</artifactId>
</dependency>
<!-- H2 Database -->
<dependency>
    <groupId>com.h2database</groupId>
    <artifactId>h2</artifactId>
    <scope>runtime</scope>
</dependency>
<!-- MySQL Driver (Optional if using MySQL) -->
<dependency>
    <groupId>com.mysql</groupId>
    <artifactId>mysql-connector-j</artifactId>
    <scope>runtime</scope>
</dependency>
<!-- iText PDF (CORRECT METHOD - NOT POM TYPE) -->
<dependency>
    <groupId>com.itextpdf</groupId>
    <artifactId>kernel</artifactId>
    <version>7.2.5</version>
</dependency>
<dependency>
    <groupId>com.itextpdf</groupId>
    <artifactId>layout</artifactId>
    <version>7.2.5</version>
</dependency>
<!-- DevTools -->
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
</dependency>
```

```
<!-- Testing -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
</project>
```